MATHEMATICS 10th (STANDARD)

Sample Question Paper 1

Time : 3 Hrs.
Max. Marks : 80

General Instructions

Read the following instructions very carefully and strictly follow them.

1. This question paper contains **38** questions. **All** questions are **compulsory**.
2. This question paper is divided into **five** Sections— **A, B, C, D** and **E**.
3. In **Section A,** Questions no. **1** to **18** are Multiple Choice Questions (MCQs) and Questions no. **19** and **20** are Assertion-Reason based questions of **1** mark each.
4. In **Section B,** Questions no. **21** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.
5. In **Section C,** Questions no. **26** to **31** are Short Answer (SA) type questions, carrying **3** marks each.
6. In **Section D,** Questions no. **32** to **35** are Long Answer (LA) type questions, carrying **5** marks each.
7. In **Section E,** Questions no. **36** to **38** are Case Study Based questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case study.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C and 2 questions in Section D.
9. Draw neat diagrams wherever required. Take $\pi = 22/7$ wherever required if not stated.
10. Use of calculator is not allowed.

Section A

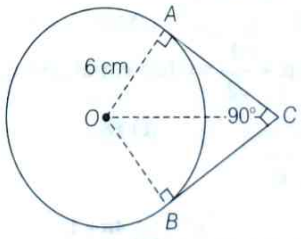
(20 × 1 = 20)

This section comprises of 18 Multiple Choice Questions and two assertion and reason type questions of 1 mark each.

1. The roots of the quadratic equation $(x+2)^2 = 25$ are

(a) 3, –7 (b) 7, –3 (c) 5, – 5 (d) –7, – 2

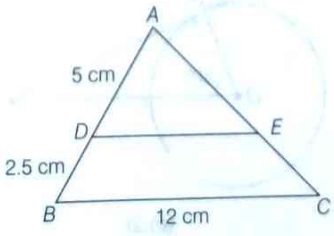
2. A circular fountain of radius 6 cm is surrounded by two straight pathways touching the fountain at points A and B. The pathways meet at point C such as in the given figure.



Find the length of each pathway from point C to the point of contact.

(a) 6 cm (b) $6\sqrt{2}$ cm (c) $3\sqrt{2}$ cm (d) 12 cm

3. In the given figure ΔABC is shown. DE is parallel to BC. If $AD = 5$ cm, $DB = 2.5$ cm and $BC = 12$ cm, then DE is equal to



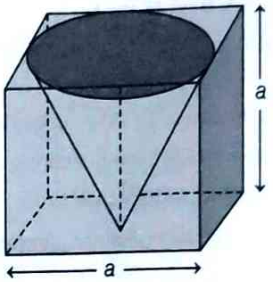
(a) 10 cm (b) 6 cm (c) 8 cm (d) 7.5 cm

4. Without carrying out complete multiplication, identify the correct property of $(5\times9\times11+5)$.
   (a) It is divisible by 5. (b) It is a prime number. (c) It is divisible by 11. (d) It is an even number.

5. Two fair dice are rolled simultaneously, then probability that the sum of the number obtained is a multiple of is
   (a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{5}{18}$ (d) $\frac{7}{36}$

6. A mathematics club is arranging numbered cards from 20 to 300. A student wants to identify the numbers that are divisible by 5. How many such numbers are there?
   (a) 55 (b) 56 (c) 57 (d) 58

7. A decorative gift box is prepared by placing a cone inside a cube. The circular base of the cone exactly fits the face of the cube and the height of the cone is equal to the edge of the cube.



   If the edge of the cube is a, what is the volume of the empty space left inside the cube?
   (a) $a^3\left(1-\frac{\pi}{12}\right)$ (b) $\frac{\pi a^3}{3}$ (c) $\frac{\pi a^3}{12}$ (d) $\frac{a^3}{3}$

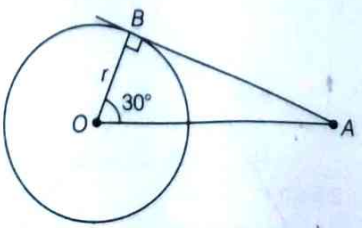
8. Rahul correctly found that two numbers are in the ratio 3 : 4 and their HCF is 7.
   Consider the following statements regarding these two numbers.
   I. The numbers are 21 and 28.
   II. The LCM of these two numbers is 84.
   III. The product of the two numbers is equal to 588.
   Which of the following statements is/are correct?
   (a) I and II only (b) II and III only (c) I and III only (d) I, II and III

9. A ladder is placed against a vertical wall. The height reached by the ladder is $\sqrt{3}$ times the distance of its foot from the wall. What is the angle made by the ladder with the ground?
   (a) 30° (b) 45° (c) 60° (d) 75°

10. If the quadratic equation $\sqrt{2}x^2+(k-2)x+\frac{9}{\sqrt{2}}=0$ has equal and real roots, then the value of k is
    (a) 8 (b) −8 (c) ±8 (d) 0

11. The nth term of the AP $\frac{5}{2},\frac{9}{2},\frac{13}{2},\frac{17}{2},\ldots$ is
    (a) $2n+\frac{1}{2}$ (b) $2n+\frac{3}{2}$ (c) $\frac{4n+1}{2}$ (d) $\frac{4n-1}{2}$

12. A circular park has centre O. A straight pathway AB touches the park at point B. A visitor standing at point A observes that $\angle AOB=30°$. If the radius of the park is r, what is the distance of OA?



    (a) $\frac{2r}{\sqrt{3}}$ (b) $\sqrt{3}r$ (c) $2r$ (d) $\frac{r}{\sqrt{3}}$

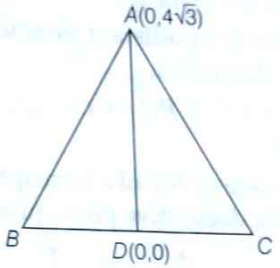
13. A scale model of a building is prepared such that $\Delta PQR \sim \Delta XYZ$. The corresponding sides satisfy $PQ = 12$ cm, $QR = 8$ cm and $XY = 3$ cm. Find the length of side YZ.
(a) 4 cm (b) 3 cm (c) 2 cm (d) 6 cm

14. A quadratic polynomial has sum of zeroes equal to 2 and product of zeroes equal to −15. This polynomial $P(x)$ is
(a) $x^2 - 2x - 15$ (b) $x^2 + 2x - 15$ (c) $x^2 - 15x + 2$ (d) $x^2 + 15x - 2$

15. Given that $\sin 2\theta = \frac{\sqrt{3}}{2}$, where $0° < \theta < 45°$. The value of $\cos 2\theta$ is
(a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) $-\frac{1}{2}$ (d) $\frac{1}{2}$

16. In a survey of marks obtained by students in a test the median and mode are 32.4 and 33.8, respectively. The mean of the distribution is
(a) 31.7 (b) 32.2 (c) 33.6 (d) 35.2

17. In the given figure, ΔABC is an equilateral triangle. AD is a median of the triangle joining the points $A(0, 4\sqrt{3})$ and $D(0, 0)$. Then, the coordinates of the points B and C (in the same order) are



(a) $(-6, 0), (6, 0)$ (b) $(-3\sqrt{3}, 0), (3\sqrt{3}, 0)$ (c) $(-4, 0), (4, 0)$ (d) $(-2\sqrt{3}, 0), (2\sqrt{3}, 0)$

18. If $4\tan\theta = 3$, then $\left(\frac{4\sin\theta - \cos\theta}{4\sin\theta + \cos\theta}\right)$ is equal to
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

Assertion and Reason Based Questions

Questions number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the options (a), (b), (c) and (d) as given below.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

19. **Assertion** (A) The length of the perpendicular drawn from the centre of a circle to a tangent is equal to the radius of the circle.
Reason (R) A tangent to a circle is perpendicular to the radius at the point of contact.

20. **Assertion** (A) $\tan 90°$ is not defined.
Reason (R) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\cos 90° = 0$.

Section B

This section comprises of 5 Very Short Answer (VSA) type questions of 2 marks each.

21. A carpenter has two wooden strips of lengths 3 m 60 cm and 4 m 80 cm. He wants to cut them into equal pieces of the greatest possible length without any wastage. Find the length of each piece.

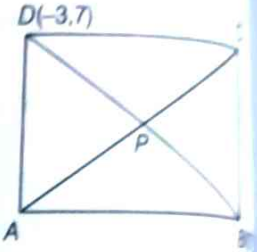
22. (a) During a plantation drive, the number of saplings planted each day forms an arithmetic progression. On the first day, 45 saplings were planted. The number of saplings planted decreased by 3 each day and on the last day 6 saplings were planted. Then, find the total number of days for which the plantation drive continued and the total number of saplings planted during the entire drive.

Or (b) The number of books arranged on successive shelves of a library forms an arithmetic progression. The number of books on the nth shelf is given by $a_n = 5n + 2$. Find the sum of the books arranged on the 25 shelves.

23. (a) Diagonals AC and BD of rectangle $ABCD$ intersect at point P. Coordinates of points B and D are $(7, 3)$ and $(-3, 7)$, respectively. (Consider line AB is parallel to X-axis).

(i) Find the coordinates of point P.

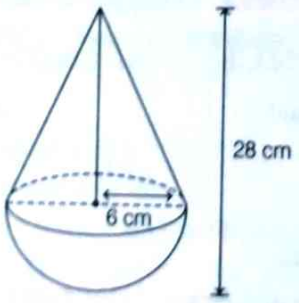
(ii) Find the length of the side AB of the rectangle.



Or (b) A point lies on the line $2x - y = 4$ and is equidistant from the points $(1, 3)$ and $(5, -1)$. Find the coordinates of that point.

24. In ΔPQR and ΔXYZ, it is given that $\angle Q = \angle Y$. If $PQ = 6$ cm, $QR = 9$ cm, $XY = 4$ cm and $YZ = 6$ cm, show that $\Delta PQR \sim \Delta XYZ$.

25. A decorative piece is made by placing a cone on top of a hemisphere of radius 6 cm. The total height of the piece is 28 cm. Find the total surface area of the decorative piece. (use $\sqrt{130} = 11.40$)



Section C

This section comprises of 6 Short Answer (SA) type questions of 3 marks each. (6 × 3 = 18)

26. (a) If $\sin\theta + \cos\theta = \sqrt{2}$, then prove that $\tan\theta + \cot\theta = 2$

Or (b) During a mathematics class, Renu wrote the expression given below on the board and asked the students to simplify it.

$(\sin A - \sec A)^2 + (\cos A - \operatorname{cosec} A)^2$

Divya solved it in her notebook as follows.

$(\sin A - \sec A)^2 + (\cos A - \operatorname{cosec} A)^2$

$= \sin^2 A + \sec^2 A - 2 \sin A \sec A + \cos^2 A + \operatorname{cosec}^2 A - 2 \cos A \operatorname{cosec} A$

$= 1 + \sec^2 A + \operatorname{cosec}^2 A - 4$ … (Step 1)

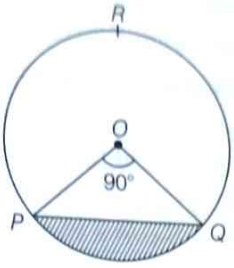
$= 1 + \sec^2 A \operatorname{cosec}^2 A - 4$ … (Step 2)

$= \sec^2 A \operatorname{cosec}^2 A - 3$ … (Step 3)

… (Step 4)

Identify the step in which Divya has made an error. Rectify the error and find the correct answer.

27. In the given figure, a chord PQ of a circle with radius 9 cm subtends an angle of 90° at the centre O. Find
(i) perimeter of major sector $OPRQ$ and
(ii) area of the shaded segment, if the area of $\Delta POQ = 36\ cm^2$.



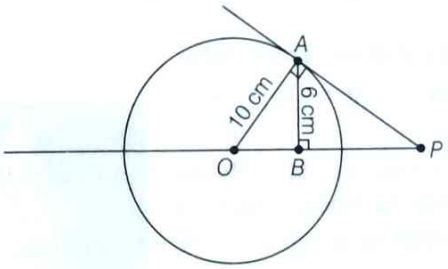
28. Prove that $5-2\sqrt{3}$ is an irrational number, it is given that $\sqrt{3}$ is an irrational number.

29. Find two consecutive negative integers, the sum of whose squares is 365.

30. The point $M(7, y)$ divides the line segment joining $A(4, -3)$ and $B(8, 5)$ in the ratio 3: 1. Find the value of y.

31. (a) Two tangents PA and PB are drawn from an external point P to a circle with centre O, touching it at A and B respectively. Prove that OP is the perpendicular bisector of AB.

Or (b) PA is a tangent to a circle with centre O such that $OA = 10$ cm, $AB = 6$ cm and $AB \perp OP$, where B lies on segment OP. Find the length of PB.



Section D

(4 × 5 = 20)

This section comprises of 4 Long Answer (LA) type questions of 5 marks each.

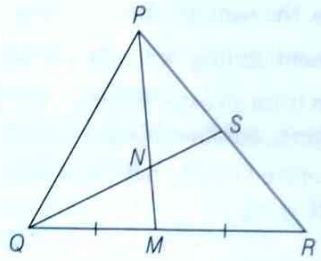
32. (a) In a fitness club, the monthly calories burnt by members are recorded in the following grouped data. The median of the data is 162. Find the values of p and q, given that the total number of members is 75.

Calories Burnt (in kcal)	**Number of Members**
80-100	5
100-120	7
120-140	p
140-160	18
160-180	25
180-200	q
200-220	3

Or (b) A survey was conducted to find the daily pocket money (in ₹) of students. Find the mean and mode of the following distribution.

Daily Pocket Money (in ₹)	Number of Students
0-20	4
20-40	8
40-60	12
60-80	16
80-100	14
100-120	7
120-140	4

33. In ΔPQR, PM is a median. N is a point on PM such that $PN:NM = 2:1$, QN is extended so that it meets PR at S. Prove that $QS = 3SN$.



34. (a) Solve the following system of equations graphically

$$2x + y = 6,\ x - 3y = -1$$

Or

(b) Six years ago, Meena was four times as old as Reena. Eight years from now, Meena will be twice as old as Reena. To know the present ages of Meena and Reena:

(i) form the linear equations representing the above information.

(ii) show that the pair of linear equations is consistent with a unique solution.

(iii) find the present age of Meena and Reena.

35. From the top of a 25 m high lighthouse, the angle of elevation of the top of a hill is 30°. A man standing at the foot of the lighthouse observes the top of the hill at an angle of elevation of 60°. If the lighthouse and the hill are on the same horizontal ground, find the horizontal distance between the foot of the lighthouse and the foot of the hill. Also, find the height of the hill from the ground. (use $\sqrt{3} = 1.73$)

Section E

This section comprises of 3 Case study/Passage-based questions of 4 marks each with sub parts. **(3 × 4 = 12)**

36. During a cultural programme, a decorative entrance arch was constructed on the stage. The shape of the arch is represented by the polynomial $p(x) = -2x^2 + 8x + 24$ where x is the distance (in feet) measured along the stage floor.



Based on the above information, answer the following questions.

(i) Determine the maximum height of the arch. (1)

(ii) The stage designer marks the point where the arch intersects the Y-axis for placing a decorative light. Write the coordinates of this point and determine its distance from the origin. (1)

(iii) (a) Find the zeroes of the polynomial $p(x)$. Which points on the graph represent these zeroes? (2)

Or (b) The distance of the centre of the arch from each supporting pillar is equal to half of the span of the arch. Determine this distance. (2)

37. A box contains 90 discs which are numbered from 1 to 90. One disc is drawn at random from the box.

Based on the above information, answer the following questions.

(i) Find the probability that the drawn disc bears a two-digit number. (1)

(ii) Find the probability that the drawn disc bears a number divisible by 5 but not divisible by 10. (1)

(iii) Based on the same box, calculate the following.

(a) Find the probability that the disc bears a perfect square number. (2)

Or (b) Find the probability that the disc bears a prime number less than 40. (2)

38. A decorative piece for a garden is in the shape of a cone mounted on a cylindrical base. The base is used as a planter and the conical part acts as a decorative top.

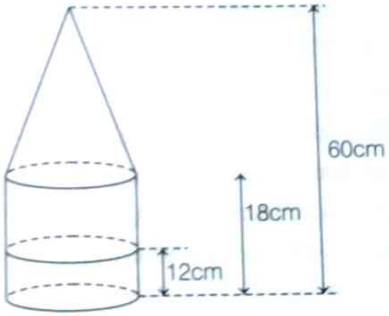
Radius of the cylindrical base is 28 cm.

Height of the cylindrical base is 18 cm.

Total height of the structure is 60 cm.

The planter is filled with soil up to a height of 12 cm.





Based on the above information, answer the following questions.

(i) Find the slant height of the conical part. (1)

(ii) Find the total surface area of the decorative piece. (1)

(iii) (a) Find the volume of the soil filled in the cylindrical planter. (2)

Or (b) Find the total volume of the decorative piece. (2)